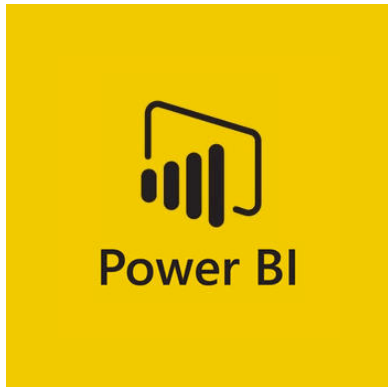




Uber Ride Analytics Dashboard Power BI Project Report

Prepared by:

Moyank Giri

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Professional Project Showcase

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Executive Summary

This project demonstrates an end-to-end **Power BI analytics solution** for Uber's NCR ride-booking data. It transforms operational records into actionable intelligence for stakeholders by visualizing booking trends, cancellations, customer feedback, and vehicle usage.

The analysis shows that while total bookings reached nearly **148.8K**, only **62%** were completed successfully. Driver cancellations and unavailability are the leading causes of inefficiency. Autos and Go Mini vehicles dominate demand, whereas premium categories remain underutilized.

This report highlights the analytical approach, dashboard design reasoning, and the insights gained, showcasing proficiency in Power BI data modeling, DAX formulation, and professional data storytelling.

1 Introduction

Ride-booking platforms like Uber rely on data analytics to optimize operations, improve customer satisfaction, and allocate resources efficiently. This Power BI project was undertaken to showcase data visualization and modeling capabilities, while deriving meaningful insights from booking datasets.

1.1 Objectives

- Design and implement a Power BI dashboard to analyze booking and cancellation trends.
- Create meaningful KPIs through DAX calculations for dynamic decision support.
- Translate raw data into business narratives for operational improvement.

1.2 Scope

The scope includes descriptive analytics for the given dataset — booking performance, cancellation breakdown, customer ratings, and vehicle utilization — through interactive and visually optimized Power BI reports.

2 Methodology and Technical Details

2.1 Data Source and Acquisition

The dataset was imported from the Kaggle repository, [Uber Ride Analytics Dashboard](#), which provides records of NCR Uber ride bookings, including:

- Booking status, timestamps, and vehicle types.
- Customer ratings and feedback.
- Cancellation details and driver behavior.

2.2 Technical Methodology

This section outlines the technical steps undertaken within the Power BI ecosystem to process, model, and analyze the Uber ride data. It covers the data transformation in Power Query, the logical structure of the data model, and the DAX formulations used to create the key analytical measures.

2.2.1 Data Preparation in Power Query

The initial raw data was processed using Power Query Editor to ensure its quality, consistency, and suitability for analysis. The following key steps were performed:

- **Data Ingestion:** The dataset was loaded into Power Query.
- **Data Profiling:** Initial assessment of column quality, value distribution, and errors was conducted to identify inconsistencies.
- **Type Correction:** Data types for each column were verified and corrected. For instance, the booking date column was set to a 'Date/Time' type, and numerical IDs were set to 'Whole Number'.
- **Data Cleansing:** Text data, such as in the 'Booking Status' and 'Vehicle Type' columns, was cleaned for consistency by trimming whitespace and standardizing capitalization.

2.2.2 DAX Formulations

Data Analysis Expressions (DAX) were used to create the calculated measures required for the visualizations. The following are the core DAX formulas used to generate the KPIs shown on the dashboard.

Total Bookings

This measure calculates the total number of ride requests in the dataset.

```
Total Bookings =  
COUNTROWS( 'Uber ' )
```

Total Cancelled Bookings

This measure filters the data to count only the bookings that were cancelled by either the customer or the driver.

```
Total Cancelled Bookings =  
CALCULATE(  
    COUNT( 'Uber ' [Booking ID] ) ,  
    'Uber ' [Booking Status] IN { "Cancelled -by- Customer" ,  
        "Cancelled -by- Driver" }  
)
```

Total Unsatisfactory Bookings

This measure calculates all bookings that were not successfully completed, providing a comprehensive view of service failures.

```
Total Unsatisfactory Bookings =  
CALCULATE(  
    COUNT( 'Uber ' [Booking ID] ) ,  
    'Uber ' [Booking Status] <> "Completed"  
)
```

Total 5 Star Bookings

This measure counts the number of rides that received a 5-star rating, assuming a 'Rating' column exists in the dataset.

```
Total 5 Star Bookings =  
CALCULATE(  
    COUNT( 'Uber ' [Booking ID] ) ,
```

```
        FILTER(  
            'Uber' ,  
            'Uber' [Rating] = 5  
        )  
    )
```

2.3 Technical Tools Used

- Microsoft Power BI Desktop (Data Model + Visualization)
- Power Query (ETL and transformation)
- DAX (Analytical measures)
- Power BI

3 Analytics and Visualization Insights

This section describes each Power BI visual, the analytical reasoning behind it, and the resulting insights.

3.1 KPI Summary Cards

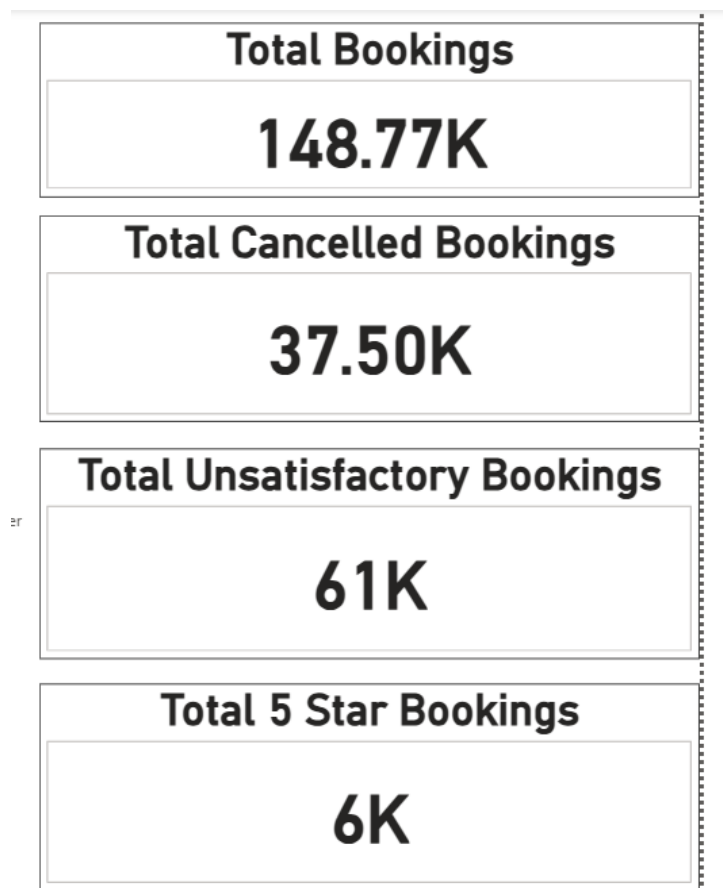


Figure 3.1: Top-Level KPIs for Ride Metrics

Analytics

This section highlights four fundamental metrics that provide a high-level snapshot of the business's health: Total Bookings, Total Cancelled Bookings, Total Unsatisfactory

Bookings, and Total 5 Star Bookings.

These KPIs serve as the primary measure of volume, service failure, and service excellence. They allow stakeholders to instantly monitor business health indicators and track targets.

Reason

KPI cards are used for their ability to convey critical business metrics at a glance. They are essential in any dashboard for providing immediate, quantifiable insights into core performance areas without needing to interpret complex charts, making them ideal for executive-level monitoring.

Key Insights

- **High Booking Volume:** The service processed a total of 148.77K booking requests over the year, indicating substantial market demand.
- **Significant Cancellation Volume:** Out of all bookings, 37.50K were cancelled by either the customer or the driver, representing approximately 25% of total requests and a major source of service failure.
- **Widespread Unsatisfactory Experiences:** A total of 61K bookings are classified as "Unsatisfactory". This figure, encompassing all non-completed rides, highlights that over a third of customer interactions do not result in a successful trip.
- **Low Rate of Top Ratings:** Only 6K bookings received a 5-star rating. When compared to the 93K completed rides, this suggests that only about 6.5% of successful trips are rated as excellent by customers.

3.2 Booking Status Distribution

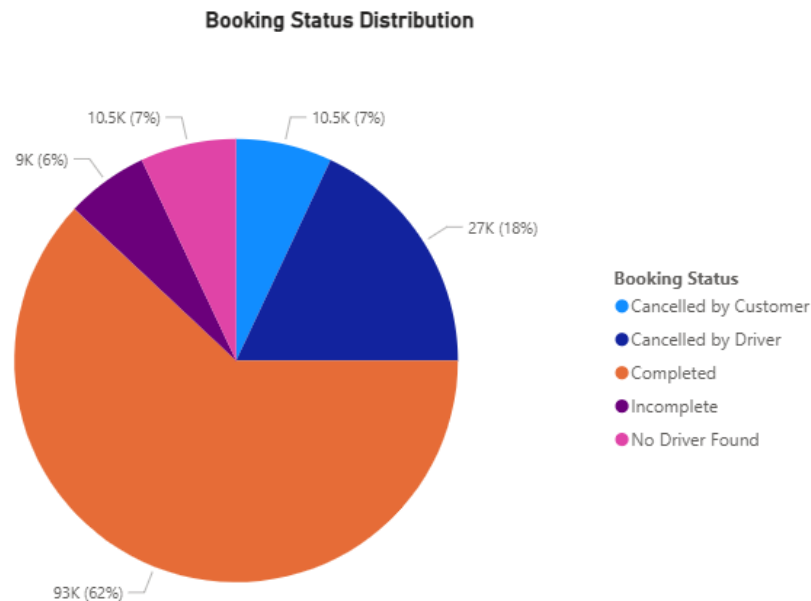


Figure 3.2: Booking Status Distribution — Power BI Pie Chart

Analytics

This pie chart provides a detailed breakdown of all booking requests, categorizing them by their final status. It quantifies the proportion of successfully completed rides against various service failures: Cancelled by Driver, Cancelled by Customer, Incomplete, and No Driver Found.

Reason

A pie chart is utilized to effectively visualize the part-to-whole relationship between each booking status and the total volume of requests. This provides an immediate, high-level understanding of the overall success rate and the magnitude of different failure points.

Key Insights

- **Low Completion Rate:** Only 62% of all bookings result in a successfully completed trip, underscoring a significant gap in service fulfillment.
- **Driver Cancellations Dominate:** Cancellations initiated by drivers (18%) are the single largest reason for service failure, occurring more than twice as often as customer-initiated cancellations (7%).

- **Driver Availability Issues:** "No Driver Found" accounts for 7% of outcomes, highlighting a significant issue with driver supply or allocation that matches the customer cancellation rate.
- **Incomplete Trips:** An additional 6% of bookings are marked as "Incomplete", representing another category of service failure that requires further investigation.

Monthly Booking Trends

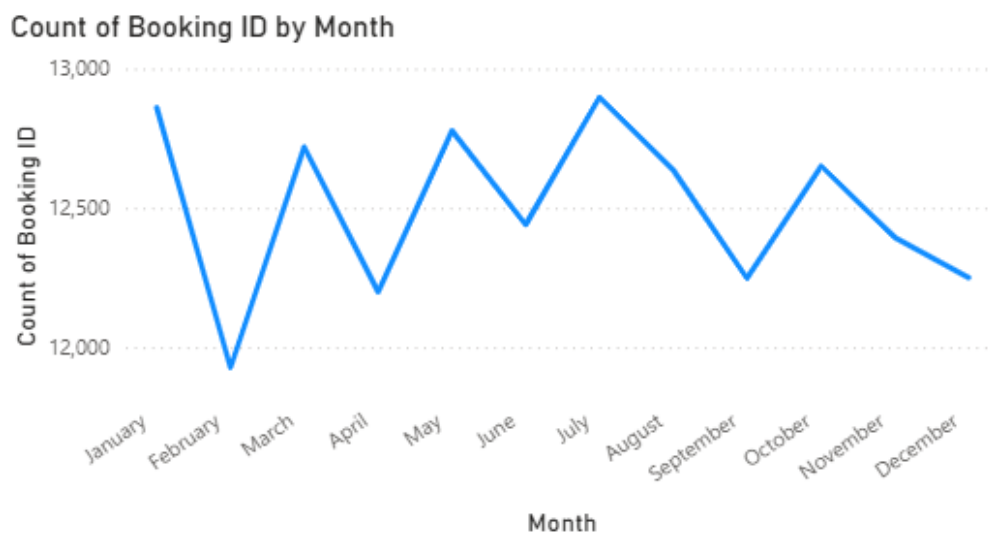


Figure 3.3: Count of Booking IDs by Month for 2024.

Analytics

This line chart illustrates the trend of total booking IDs on a monthly basis throughout 2024. It is used to identify patterns, seasonality, and the overall stability of demand over the year.

Reason

A line chart is the ideal visualization for time-series data. It clearly connects consecutive data points (months), making it easy to spot trends, fluctuations, and patterns over a continuous period.

Key Insights

- **Stable Demand:** The volume of bookings remained relatively consistent across the year, fluctuating in a narrow band between approximately 12,000 and 13,000 bookings per month.

- **No Strong Seasonality:** The data does not indicate any significant seasonal spikes or dips, suggesting a mature market with predictable, year-round demand.
- **January Peak:** The year began with the highest booking volume, reaching around 12,500 in January, followed by minor fluctuations in subsequent months.

3.3 Vehicle Type Utilization

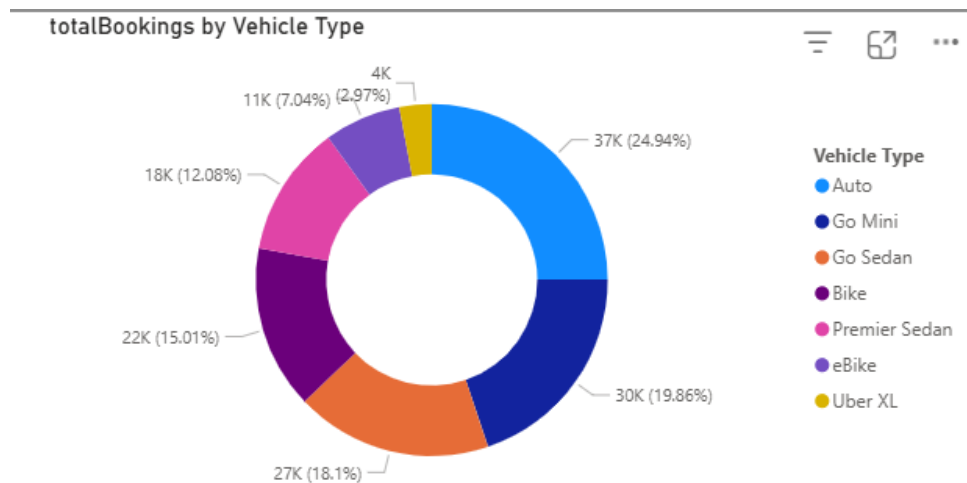


Figure 3.4: Vehicle Type Utilization — Power BI Bar Chart

Analytics

This bar chart compares the total number of bookings across Uber's different vehicle service tiers. It highlights customer preferences and which services contribute most to the overall booking volume.

Reason

A bar chart is effective for comparing the magnitude of different discrete categories. In this case, it allows for a straightforward comparison of the popularity of each vehicle type, making it easy to identify the most and least requested services.

Key Insights

- **Top Tier Services:** Go Sedan is the most requested service with 37K bookings (24.94%). Go Mini (30K) and Auto (27K) are also highly popular, and together these three categories account for over 60% of all rides.
- **Mid-Tier Preference:** Premier Sedan (22K) and Bike (18K) represent a secondary but significant portion of the demand.

- **Niche Services:** Uber XL (11K) and eBike (4K) are the least popular options, suggesting they cater to a smaller, more specific segment of the user base.

3.4 Analysis of Customer Cancellation Reasons



Figure 3.5: Top Reasons for Ride Cancellation by Customers.

Analytics

This visualization lists the primary reasons cited by customers when they cancel a ride. This qualitative data provides direct insight into the specific friction points in the user experience that lead to a customer aborting a booking request.

Reason

A categorized list is a simple yet powerful way to present qualitative feedback. It directly surfaces the voice of the customer, allowing stakeholders to understand the "why" behind the cancellation metrics and prioritize fixes for the most pressing issues.

Key Insights

- **Poor Driver Experience:** Reasons such as "Driver is not moving towards pickup location" and "Driver asked to cancel" point directly to negative driver behavior and service quality issues as key drivers of customer cancellations.
- **Customer-Side Factors:** The reason "Change of plans" indicates that not all cancellations are due to service failures, but also due to external customer factors.
- **Platform/Logistical Issues:** "Wrong Address" suggests potential issues with the app's location services or user error during booking, representing an opportunity for UX improvement.

4 Recommendations

Based on the analysis, the following actions are recommended to improve service performance:

1. **Address Driver-Side Service Failures:** With driver cancellations at 18% and customer complaints about driver behavior (e.g., "Driver asked to cancel"), it is critical to focus on driver satisfaction and accountability.
 - **Action:** Implement a driver incentive program that rewards low cancellation rates. Investigate the root causes of cancellations through driver surveys, focusing on fare structure, trip distance, and app functionality.
2. **Optimize Driver Supply and Dispatch:** A 7% "No Driver Found" rate indicates a mismatch between driver supply and rider demand in certain areas or times.
 - **Action:** Utilize geospatial analysis to identify service gaps. Introduce targeted "surge" incentives for drivers to operate in these specific zones and improve the efficiency of the driver dispatch algorithm.
3. **Enhance Customer Experience and Ride Quality:** The high number of unsatisfactory bookings (61K) and low proportion of 5-star ratings (6K) show a clear need for quality improvement.
 - **Action:** Address the technical and logistical issues cited by customers, such as improving address accuracy in-app to reduce "Wrong Address" cancellations. Monitor metrics like "Driver is not moving" to flag and train or penalize non-compliant drivers.
4. **Focus on Core Vehicle Categories:** Go Sedan, Go Mini, and Auto are the clear market leaders.
 - **Action:** Concentrate marketing campaigns and promotions on these core services to maximize return on investment. For low-performing services like eBike, conduct a profitability analysis to determine if the service should be optimized or discontinued.

5 Conclusion

This project demonstrates how Power BI can transform operational ride data into strategic insights through careful data modeling, DAX measure formulation, and storytelling-driven visualization. By analyzing patterns in cancellations, vehicle demand, and customer ratings, Uber's NCR operations can enhance completion rates, improve satisfaction, and optimize resource deployment.

The report also illustrates end-to-end analytics competency—from data ingestion to dashboard design—serving as a professional portfolio piece highlighting Power BI expertise, business understanding, and result-oriented insight delivery.